ICT PROJECT / STAGE 2

Campus Lost & Found

Working local prototype
Endterm demonstration

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Synthetic data. Local authentication.
University SSO remains pending.

Student workflow

LIVE DEMO / 70 SECONDS

01 Publish a found notice

Enter details and attach one JPG or PNG, up to 5 MB.

02 Find the saved notice

Search its title and choose a category.

03 Contact the author

Open the university email link. No message is sent in the demo.

04 Edit and mark returned

The owner controls changes. SQLite keeps the saved status.

Administrator flow: review misuse, confirm removal.

Requirements and access

01 Core notice workflow works

F2, F4–F8: publish, browse, search, contact, edit, return and moderate.

02 Private photo upload works

F3: one JPG or PNG up to 5 MB. The server validates and re-encodes it.

03 Owner and administrator checks

The server blocks forbidden actions and protects state changes with CSRF checks.

04 University identity remains a limit

F1: local registration restricts the domain. University SSO and account ownership verification remain pending.

The demonstration runs over local loopback HTTP.

Test evidence and open checks

81.47 ms

Observed search p95 against a 2,000 ms target

10 concurrent sessions, 100 local HTTP requests.
Asset loading and browser painting are excluded.

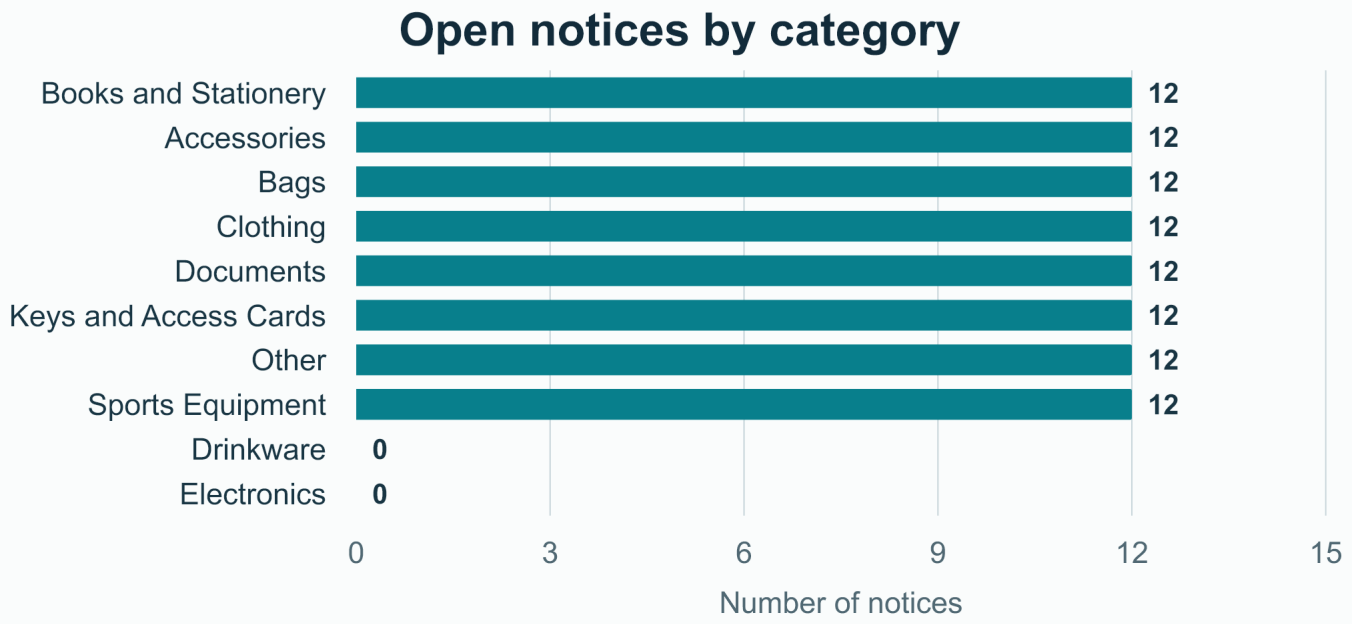
Original success criterion	Current evidence
4/5 testers post within 2 minutes	PENDING
4/5 testers find within 1 minute	PENDING
5/5 invalid-access checks blocked	5/5 blocked

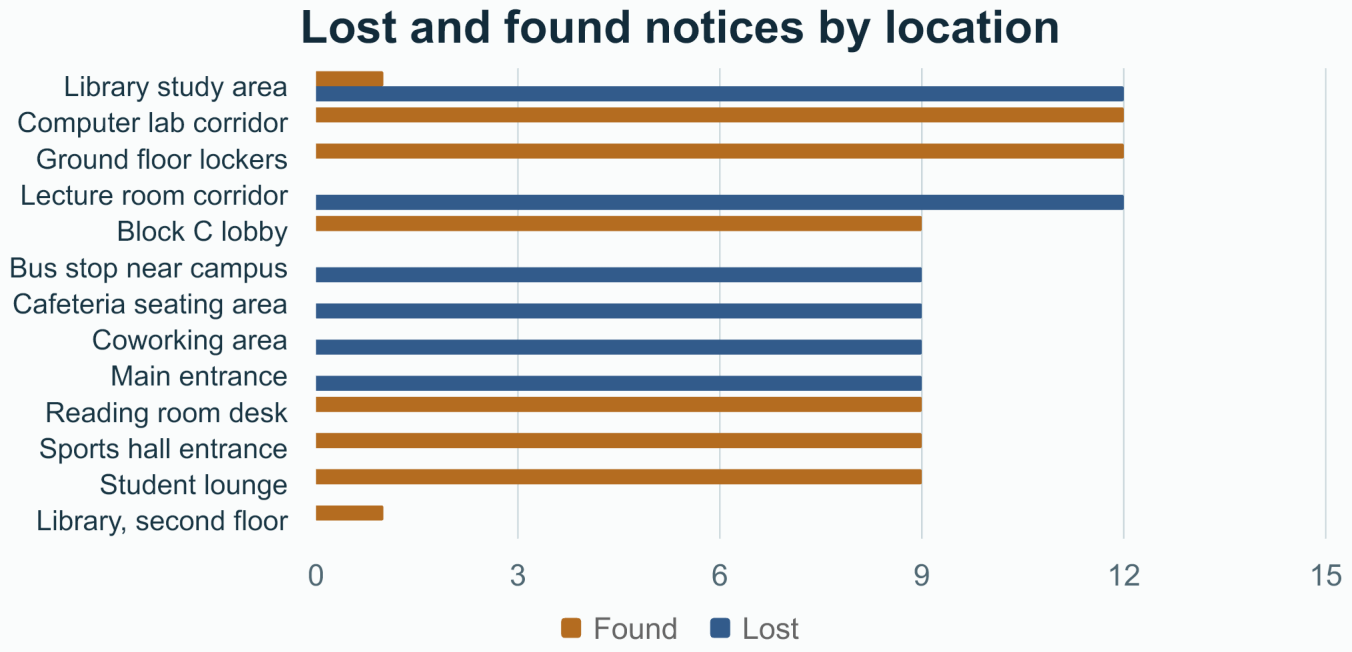
HTTP availability: 20/20 across 30 minutes.
N4: PARTLY — offline/retry still pending.

Pending: five outside testers, three observed usability problems, current browser checks and three Fast4G page loads.

Category and location patterns

122 synthetic notices, frozen 30 Oct 2026 at 10:34 UTC

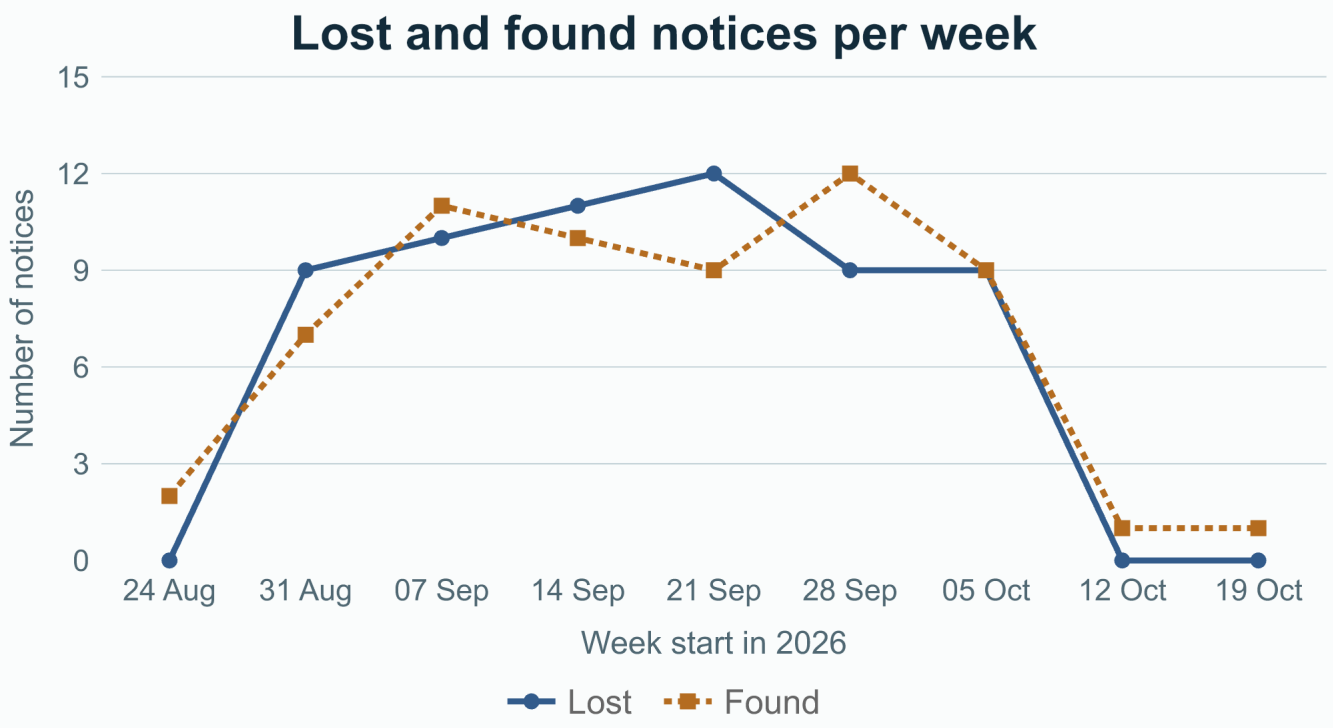




Seed bias: all 12 Electronics and 13 Drinkware notices are returned. These are demonstration patterns.

Weekly analysis and a design decision

Item event dates, grouped into Monday–Sunday weeks



Decision: standardise location choices

Free-form labels split equivalent places. Add a consistent location list, then review it with users.

LIVE SQL / 20 SECONDS

Run the category query and verify the open counts.

IoT at the University of Birmingham

23,000

Enlighted sensors in lighting fixtures

Energy use, occupancy, movement and temperature



Responsive lighting and room-use planning

Privacy risk: linked occupancy histories may reveal routines. Restrict access and retention.

Birmingham's exact hosting option is not specified in the cited campus sources.

Proposed smart-locker extension

FOUR-LAYER DESIGN

1 / Device	Door switch and Wi-Fi controller
↓	
2 / Network	Physical pilot: IoT Wi-Fi and HTTPS
↓	
3 / Processing	Backend validates notices.id and stores telemetry
↓	
4 / Application	Staff dashboard shows state and warnings

Physical proposal: open longer than 30 seconds

LIVE SIMULATION: 10 seconds, then refresh

Planning estimate: 10 lockers, one heartbeat per minute plus state changes ≈ 14,600 records or 2.92 MB per day.

Simulated readings. No physical controller or automatic return.

Before submission and Week 10

01 Before submission / external tests

Five outside testers; timed tasks; three real usability findings.

02 Before submission / browser checks

Capture 360 px, keyboard and offline retry evidence, plus three Fast4G loads.

03 Week 10 / fix and retest

Use the recorded findings to improve the interface and repeat affected checks.

04 Week 10 / rehearse and prepare

Rehearse the team defense and plan university SSO, HTTPS and privacy controls.

Stage 2 evidence is due in Week 9.